# Problem set 2

*Twists and Wrenches*

Modern Robotics

2013

## 1. $\dot{H}$

Suppose that you want to use a homogeneous matrix $\mathrm{H}_\mathrm{b}^0(t) \in SE(3)$ to describe the orientation of a body (frame) $\Psi_\mathrm{b}$ with respect to some inertial frame $\Psi_0$—for instance in a simulation. Of course, in the simulation you can calculate the velocity of a body easily enough by integrating the force, using $\dot{v} = 1/m \cdot F \Rightarrow v = 1/m \int F$.[1] Doing this in a three-dimensional Euclidean space will give you the full 6-D twist.

Now for the question: how can you calculate $\mathrm{H}_\mathrm{b}^0(t)$, given $\mathrm{H}_\mathrm{b}^0(0)$ (the initial configuration) and the twist of the body expressed in its own frame, $\mathrm{T}_\mathrm{b}^{\mathrm{b},0}$?

*Hint: consider the definition of the tilde form of the twist $\tilde{T}_b^{b,0}$.*

## 2. Tilde form of twists

Show that the left or right translation of velocity vectors $\dot{H}$ (for $H \in SE(3)$), that result in $\tilde{\mathrm{T}}_\mathrm{i}^{\mathrm{i,j}}$ and $\tilde{\mathrm{T}}_\mathrm{i}^{\mathrm{j,j}}$ respectively, indeed belong to

$$se(3) := \left\{ \begin{pmatrix} \Omega & v \\ \mathbf{0} & 0 \end{pmatrix} \text{ s.t. } \Omega \in so(3),\ v \in \mathbb{R}^3 \right\} \tag{1}$$

*Hint: look at how the right and left translations are defined.*

## 3. Fixed frames

Consider the two bodies of Figure 1, which move with respect to each other and both have two coordinate frames affixed to it. You can describe the relative motions of these two objects in various ways: $\mathrm{T}_3^{1,1}, \mathrm{T}_4^{2,2}, \mathrm{T}_4^{1,1}$ for example all represent the motion of body B with respect to body A, expressed in one of the frames on A.

Try tro prove that $\mathrm{T}_3^{2,2}$ is equal to $\mathrm{T}_4^{2,2}$.[2]

*Hint: use the tilde form of the twists; and perhaps start by showing that $T_4^{3,3}$ is zero.*

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[1] You might know this relation as $F = ma$ instead.

[2] Both represent the motion of B w.r.t. A expressed in the same frame, so the result should be the same indeed.

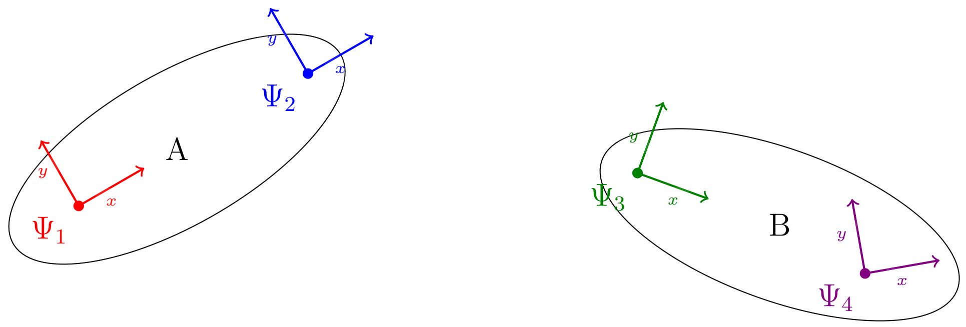


Figure 1: Two bodies with both two frames affixed to it.

## 4. Earth and Sun

Let's look at a simple example to apply the theory of twists and H-matrices. Consider a simplified model of the Sun and the Earth, where the Earth is rotating around the Sun in a perfect circle, as if it were fixed to the sun with a rod (see Figure 2).

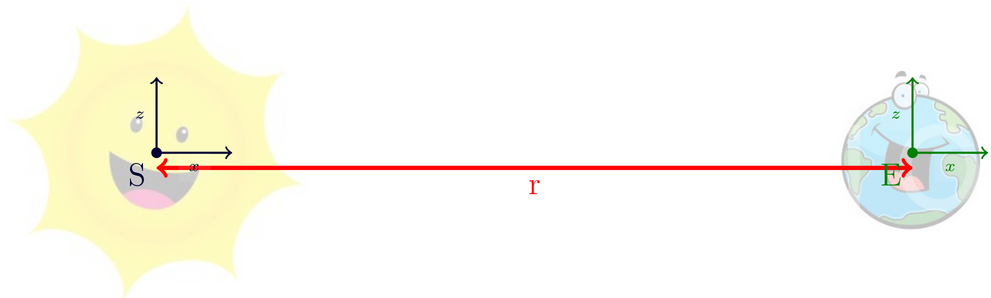


Figure 2: The Sun and the Earth.

Suppose that the motion of the Earth around the Sun is a pure rotation with a velocity of $1\,\mathrm{rad\,s^{-1}}$ around $\hat{z}$, as seen from the Sun. Now say that the Earth is located $2\,\mathrm{m}$ from the Sun[3].

We Earthlings are interested in the speed *we* see, i.e. the same motion but then expressed in *our* frame, so please calculate this velocity.

*Hint: Of course you express all velocities as twists, so that you can use H-matrices and Adjoints to express these motions in different frames.*

[3]It is a shrunken-down, sped-up version of the solar system